

Agents for Inclusive Science Communication

Science Journal



Notes: Image generated with ChatGPT. Prompt: "Create an image about SoCal beaches"

Module 1.1: How is Our Ocean Changing?

Student Reflection

Now that you have explored how Southern California's ocean is changing due to climate change. Take a moment to reflect on your experiences during Module 1.1.

1. How has your understanding of how climate change affects Southern California's ocean changed after completing Module 1.1?

The goal by the end of the project is to construct a model of how climate change is impacting Southern California's ocean.

1. What information are you missing that can help you construct a model of how climate change is impacting Southern California's ocean?
2. What other sources or people could help you learn more information about what you want to know?

Module 1.2: Planning Community Interviews

Student Task: Interview Planning

During this module, with your research team, you created an interview plan to gain more information on how climate change will impact Southern California’s ocean.

Select 3-4 interview questions to ask your community members. You can:

- Use one of the examples below, or
- Create your own using one of the sentence stems as a guide.

Personal Connection: ● What experiences have you had with the ocean? ● Can you share a memorable...?
Climate Impact Questions: ● How has climate change affected your community? ● In what ways has your work/lifestyle/culture been influenced by...?
Human Role & Future Questions: ● What do you wish people understood about climate change and the ocean? ● What do you think will happen...?

Write the questions that you select or create below:

Question 1:

Question 2:

Question 3:

Question 4:

Module 1.2

Part 1: Planning Community Interviews

Student Task: Interview Etiquette

During this module, you reflected on the differences between effective and ineffective interview etiquette.

Read each short excerpt of an interview with a community member. Decide whether the interviewer used effective or ineffective interview etiquette, and explain your reasoning.

Excerpt 1:

Interviewer: “Have you noticed changes in the ocean over time?”

Interviewee: “Yes, actually, I’ve seen more algae blooms and—”

Interviewer: “Oh yeah! I read about that! Did you also see dolphins around here?”

1. Was this effective etiquette? ☐ Yes ☐ No

2. What made this etiquette effective or ineffective? Why?

3. If the etiquette is ineffective, how would you improve the interviewer’s behavior or approach?

Excerpt 2:

Interviewer: (smiles) "Thanks again for speaking with me today. I'm really excited to hear your thoughts."

Interviewee: "Of course!"

Interviewer: (leans in slightly, maintains eye contact) "Can you tell me what changes you've noticed in the water temperature over the years?"

1. Was this effective etiquette? ☐ Yes ☐ No

2. What made this etiquette effective or ineffective? Why?

3. If the etiquette is ineffective, how would you improve the interviewer's behavior or approach?

Excerpt 3:

Interviewer: "Don't you think pollution is the biggest reason the ocean is getting worse?"

Interviewee: "Well... I guess that's part of it, but there are other factors too."

1. Was this effective etiquette? ☐ Yes ☐ No

2. What made this etiquette effective or ineffective? Why?

3. If the etiquette is ineffective, how would you improve the interviewer's behavior or approach?

Module 1.2

Part 1: Planning your Community Interviews

Student Reflection

During this module, you identified an individual to interview and prepared interview questions to gain more information on how climate change will impact Southern California's ocean. Before you conduct your interview, take a moment to reflect on your experiences during module 1.2.

1. What did you learn about planning and preparing for your community interview? How has your thinking about asking questions or gathering information changed?
2. What do you hope to learn from your community interview?
3. What questions do you still have before conducting your interview? How could you find answers or prepare?

Module 1.2

Part 2: Conducting Community Interviews

Student Task: Optional Interview Notes

Name of Interviewee:

Date and Time of Interview:

Question:

Answer:

Question:

Answer:

Question:

Answer:

Question:

Answer:

Module 1.3: Analyzing Interview Data

Student Reflection

During this module, you analyzed your interview responses and thought about why people value the ocean. Take a moment to reflect on what you learned in Module 1.3.

1. What did you learn from analyzing your interview responses? How has your thinking about why people value the ocean or how it is changing been affected?
2. Why is it important to analyze responses to interview questions? What can this tell us about how communities value the ocean?
3. How will you use what you learned in Module 1 to build your model of how climate change is impacting Southern California's ocean? For example, what information will you add or show to explain these impacts?

Module 2.1: Building a Model

Student Task: Brainstorming Components, Processes, & Interactions

*With your team, brainstorm the **components** and **processes** that make up the ocean and add your responses to the components and processes sections below.*

Afterwards, brainstorm the **interactions humans** have with the ocean system and add your responses to the interactions section below.

- **Components:** Individual elements that make up the ocean system, which can include both living (biotic) and non-living (abiotic) elements, such as organisms like fish or kelp, water, oxygen, and nutrients.
- **Processes:** Actions in the ocean system that occur over time, such as tidal shifts, storm events, or photosynthesis.
- **Interactions:** How humans and different components of the ocean system engage with one another.
 - i. Add in parentheses () whether these interactions could be positive, negative, or neutral. Examples of interactions include beach clean-ups, emitting carbon, or surfing.

Components	Processes	Interactions
<i>shellfish</i> <i>trash</i>	<i>photosynthesis</i> <i>severe storm</i>	<i>Emitting carbon from</i> <i>factories (negative)</i> <i>Beach clean-ups (positive)</i> <i>Coastal tourism (varied)</i>

*The next step is to **select a value** your team wants to investigate in your model. Think back to your community interviews and interview analysis and answer the following with your team:*

1. Based on your interview responses, what evidence shows why people care about the ocean? Include specific quotes or examples from your interview to support your answer (reference [Module 1.2 Part 2: Conducting Community Interviews notes](#)).

2. How did your team categorize these values or themes?

3. Which value from your interviews will your team include in your model, and how does this value connect to the ocean system or the impacts of climate change?

*Now that you have chosen a value to focus your model around, it is time to refine your brainstorming list. With your team, choose and **highlight the 10-12 components, processes, and interactions** from your list (previous page) that are **most connected** to your chosen value.*

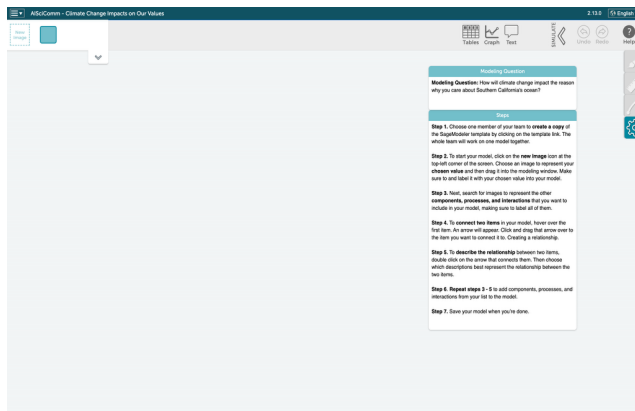
Module 2.1: Building a Model

SageModeler Instructions

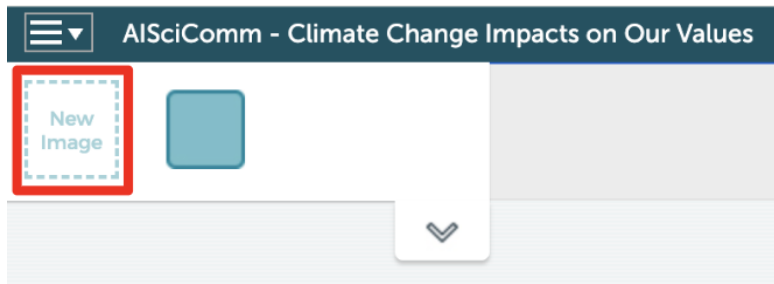
Follow these steps with your team to create your model on how climate change impacts your chosen value.

1. Choose one member of your team to **create a copy** of the [SageModeler Template](https://tinyurl.com/AISciModel) or visit <https://tinyurl.com/AISciModel>.

The whole team will work on one model together.

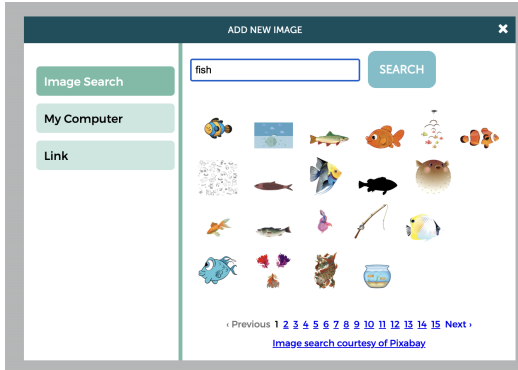


2. To start your model, click on the new image icon at the top-left corner of the screen.

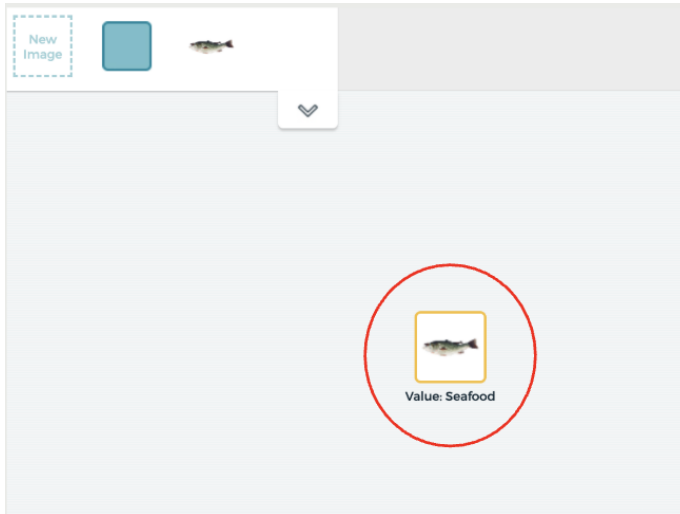


Choose an image to represent your chosen value. Press “Add Image.”

Example: Search “fish” for the value seafood.



Drag the image in the modeling window. Make sure to **label it** with your **chosen value** in your model.



3. Search for images to represent the other **components, processes, and interactions** that you want to include in your model. **Label all of them.**

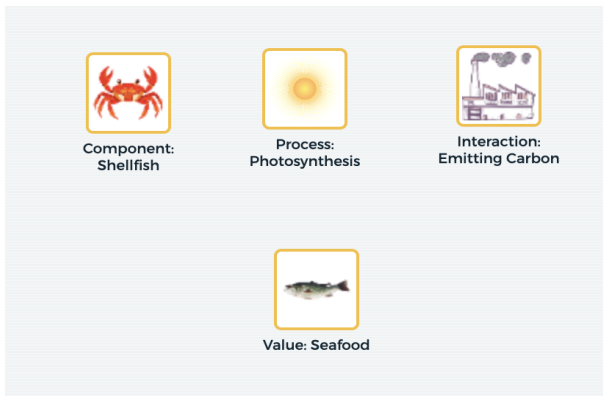
Example:



4. To **connect two items** in your model, hover over the first item. An arrow will appear.



Click and drag that arrow over the item you want to connect it to. Creating a causal relationship.



5. To **describe the relationship** between two items, double-click on the arrow that connects them.
 - a. **Positive relationship:** As one factor increases, the second factor also increases.

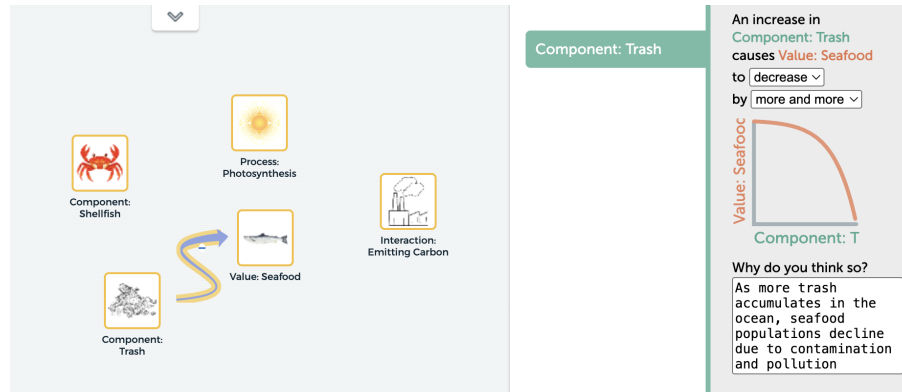
Component: Shellfish

An increase in
Component: Shellfish
causes Value: Seafood
to increase
by more and more

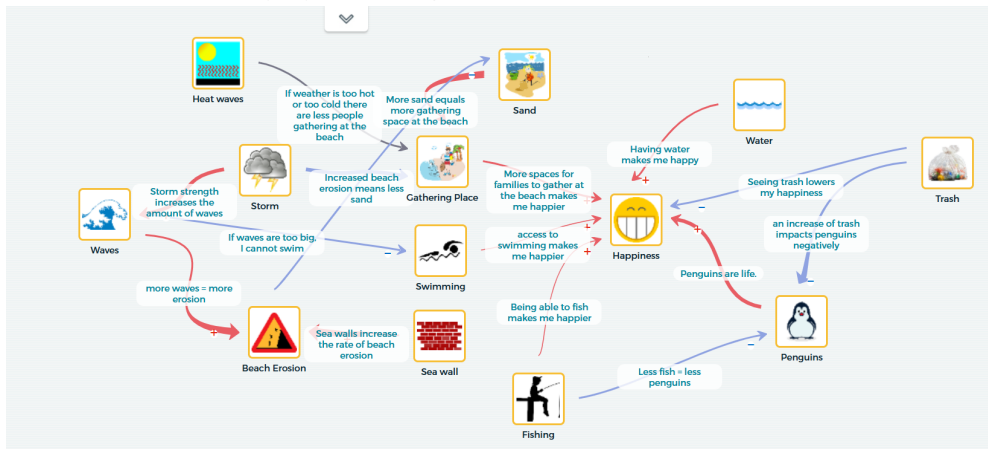
Why do you think so?

As shellfish increase more and more, there is more and more seafood available.

- b. **Negative relationship:** As one factor increases, the second factor decreases.



- c. **Varied relationship:** As one factor increases, the second factor can increase or decrease depending on different conditions.
- d. **OPTIONAL:** To make sense of the various interactions, you may add a brief explanation for the relationships (positive, negative, varied).



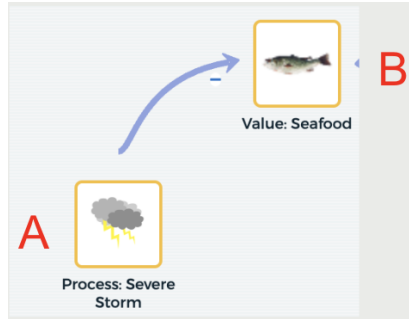
6. There are different types of causal relationships that you can model. Try to model different types of causal relationships.

- a. **Linear Chain (A → B)**

Concept: A direct cause-and-effect relationship where one element or process directly impacts another.

Example:

A heavy storm (A) physically prevents fishing vessels from leaving port, directly reducing the immediate supply of fresh seafood (B).

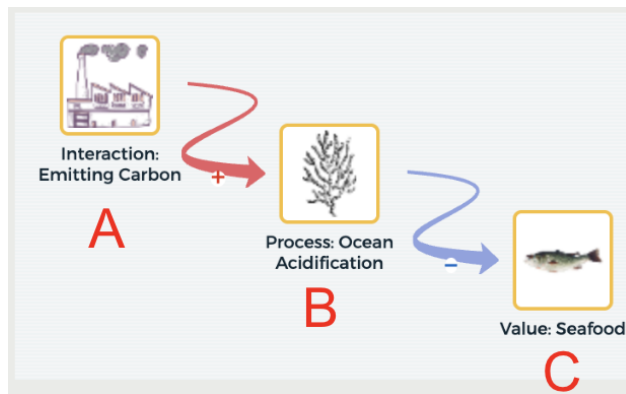


b. **Causal Chain (A → B → C)**

Concept: A multi-step sequence where an initial event triggers an intermediate change, which then leads to a final outcome.

Example:

When factories release carbon dioxide into the air (A), the ocean absorbs it and becomes acidic (B), dissolving oyster shells and reducing seafood yields for farmers (C).

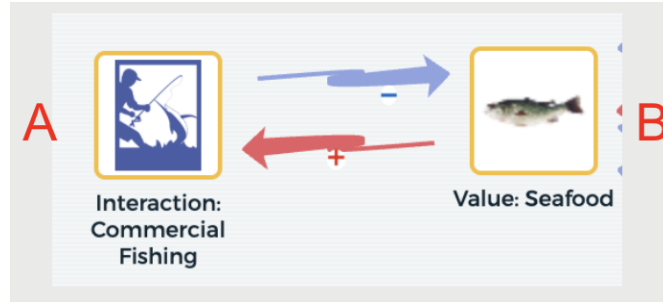


c. **Feedback Loop (A → B; B → A)**

Concept: A circular relationship where the outcome of a process feeds back into the original cause, amplifying (positive feedback) or reducing (negative feedback) the effect over time.

Example:

Intense fishing (A) drives down wild fish populations (B), creating market scarcity that raises prices and incentivizes companies to fish even harder (A), driving stocks into a deeper decline.



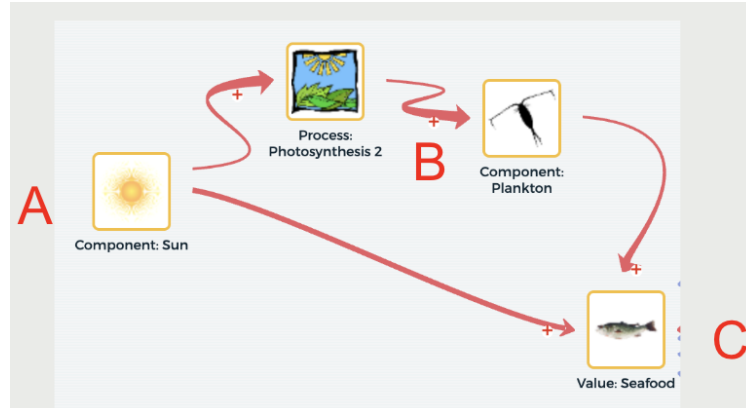
d. **Indirect Link ($A \rightarrow B$; $B \rightarrow C$; $A \rightarrow C$)**

Concept: A variable (A) influences an outcome (C) both indirectly through an intermediate variable (B) and through a separate direct pathway.

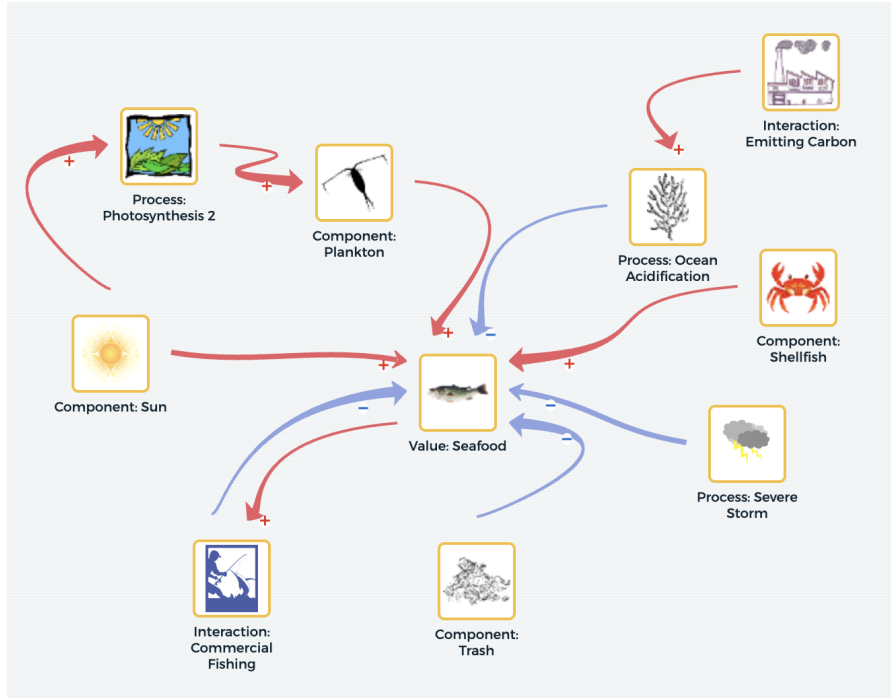
Example:

Indirect Pathway ($A \rightarrow B$; $B \rightarrow C$): Increased sunlight boosts phytoplankton growth ($A \rightarrow B$), which feeds small marine life and ultimately builds up commercial fish populations ($B \rightarrow C$).

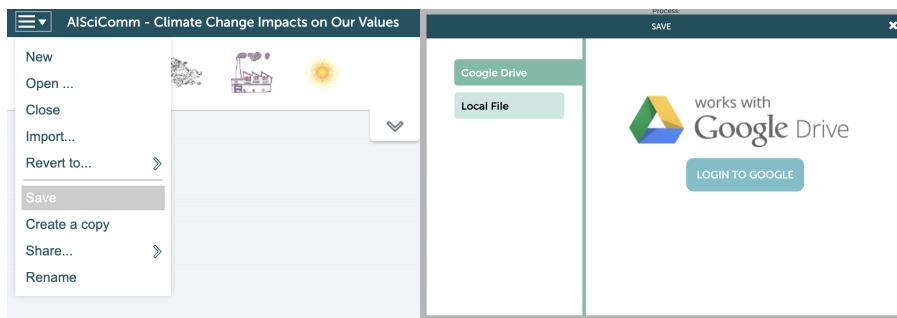
Direct Pathway ($A \rightarrow C$): Sunlight directly warms surface waters and alters fish migration routes ($A \rightarrow C$), changing where fish gather and affecting local seafood availability and prices.



- Continue to add components, processes, and interactions from your list to the model. Create different causal relationships- linear chains, causal chains, feedback loops, and indirect links.

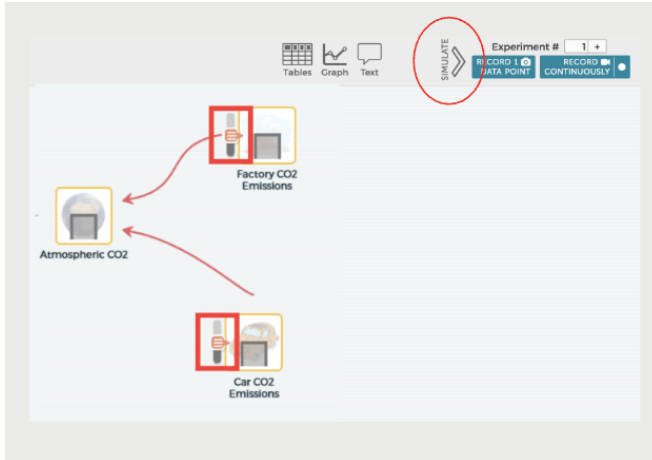


- Save your model when you're done. Connect to Google Drive to save your model in a shareable location.



- Now that your team has a complete model, take a moment to simulate change to see how changes in a portion of your model impact your chosen value.

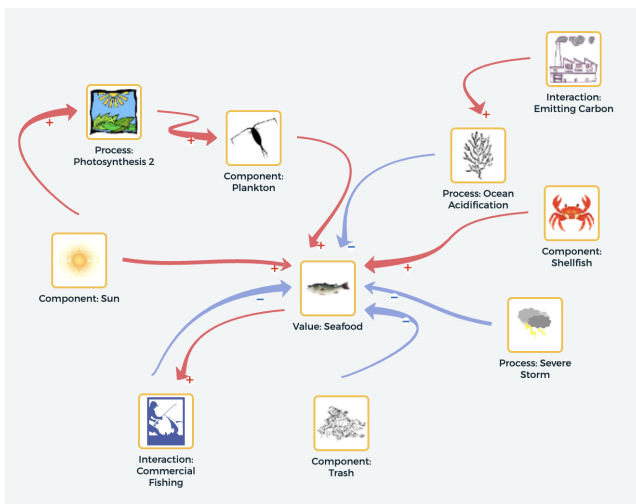
Click on the **Simulate** icon in the upper right-hand corner of SageModeler to simulate change on a different factor within your model.



As you test your simulations, think about the following:

How does a change in a component, process, or interaction within your model affect your team's chosen value?

10. Now that your team has completed their model, **take a screenshot and save your chatbot conversation** in your Google Drive. Save this snapshot as you will need it for Module 4.4.



- a. Devices will differ in ways to take a snapshot.
 - i. If on a Windows computer
 1. Press **Window + Shift + S** at the same time
 - ii. If on a Mac computer
 1. Press **Command (⌘) + Shift + 4** at the same time
 - iii. If on a Chromebook
 1. Press **Ctrl + Shift + Show Windows** at the same time

Module 2.1: Building a Model

Student Reflection

During this module, you identified the components, processes, and interactions that make up the ocean and selected a value to focus your model around. Take a moment to reflect on your experiences during module 2.1.

1. What did you learn about the components, processes, and interactions in the ocean while building your model?
2. How are these components, processes, and interactions connected to the value you chose?
3. Which parts of your model would you like to learn more about? How might you investigate or gather information to explore these areas further?

Module 2.2: Refining Your Model with GenAI

Student Task: Preparing Your Chatbot Interviews

Take a moment to explore the AISciComm chatbots to see which chatbots are available and what types of information they can provide.

With your team, you will be “interviewing” the Chatbots, similar to how you interviewed a community member in Module 1. Follow these steps to prepare your interview with the chatbots and fill in the table on the following page.

1. **Identify your focus: What questions do you have about your model?**
Look at your model. What parts do you want to learn more about?
2. **Choose a chatbot: Which Chatbot can help you gather more information about your question?**
Pick 1-2 chatbots that might help answer your question.
3. **Ask and note: After talking to the chatbot, what do you want to add to your model?**
Ask your question and write down the chatbot’s main points.

Example:

What questions do you have about your model?	Which Chatbot can help you gather more information about your question?	After talking to the chatbots, what do you want to add to your model?
What can I measure to understand how ocean acidification affects fish populations?	☐ Civil Engineer ☐ Influencer ☐ Student ☐ Kelp Researcher ☐ Fisherman	I would like to add: ● Temperature rise ● Sea level rise ● Ocean acidification ● Biodiversity loss

Preparing Your Chatbot Interviews

With your team, fill out the table below to help you plan your chatbot interviews.

What questions do you have about your model?	Which Chatbot can help you gather more information about your question?	After talking to the chatbots, what do you want to add to your model?
	☐ Civil Engineer ☐ Influencer ☐ Student ☐ Kelp Researcher ☐ Fisherman	
	☐ Civil Engineer ☐ Influencer ☐ Student ☐ Kelp Researcher ☐ Fisherman	
	☐ Civil Engineer ☐ Influencer ☐ Student ☐ Kelp Researcher ☐ Fisherman	

Module 2.2: Refining Your Model with GenAI

Student Reflection

During this module, your team explored areas in your model that you wanted to learn more about and “interviewed” the AISciComm Chatbots. Then you refined your model using the feedback provided. Take a moment to reflect on your experiences during module 2.2.

1. How did the AI chatbots help change your thinking and support the changes in your model?
2. Which areas of your model need more information? How might you explore them further?

Module 2.2: Refining Your Model with GenAI

Optional: Reflecting on AI's Responses to Your Questions

With your team, take a moment to reflect on the questions your team asked the AISciComm Chatbots during Module 2.2 Student Task: Refining Your Model with GenAI and answer the following questions:

1. Did your team ask questions that followed the CLEAR framework (Concise, Logical, Explicit, Adaptive, Reflective)? How did your questions reflect this? If not, why?
2. Did the chatbots provide answers that helped you learn more about your question? What kind of information or examples did they give? How did this connect to your community, social networks, local knowledge, or experiences?
3. Were the chatbot responses relevant to your question? Did you notice any inaccuracies or missing information? Give specific examples.

4. Generative AI chatbots learn by adjusting patterns from the data they are trained on, while traditional search engines retrieve existing information. How does understanding this difference change the way you use the chatbots? How did this influence your model updates?
5. What ideas would you give to improve the chatbots' answers? How could the chatbots better support your learning about climate change and the ocean, while reflecting your community context and assets?

Module 3.1: The Carbon Cycle

Student Task: Create a Model of the Carbon Cycle

Build a model of the carbon cycle to show how carbon moves through Earth's systems, how humans affect it, and how these changes impact your community and the Southern California ocean. Choose a process you would like to explore further.

Part 1: Label Earth's Spheres

Label the spheres on your baseline carbon cycle model. Include the following Earth Spheres:

- ☐ Atmosphere (air & gas)
- ☐ Hydrosphere (oceans, rivers, lakes)
- ☐ Biosphere (plants and animals)
- ☐ Lithosphere (land, rocks, fossil fuels)

Part 2: Human Contributions

*Use a **different color pen or sticky note** to add human-related carbon sources to your models.*

Ideas may include:

- *Fossil fuels (cars, electricity, industrial factories)*
- *Deforestation (cutting down trees)*
- *Agriculture and livestock*

Part 3: Community Connections

Look at the models you built in Module 2. Reflect on how the carbon cycle affects the ocean and your community. Use your model to make connections.

1. What do you or your community value about the Southern California ocean?
2. How might rising CO₂ levels or changes in the carbon cycle affect those values?

Part 4: Choosing a Process to Investigate

Think about the values you identified for your community. Which natural process connected to the carbon cycle do you want to explore further?

- ☐ Sea level rise
- ☐ Coastal erosion
- ☐ Ocean acidification and food webs

Reflect:

1. Which process do you want to learn more about? Why did you choose this process?
2. How does this process connect to the carbon cycle and your community?

Module 3.1: The Carbon Cycle

Student Reflection

During this lesson, you explored the natural carbon cycle, added human influences, and connected these ideas to what your community values about the Southern California ocean. You have also identified a natural process that you would like to investigate further. Take a moment to reflect on your experiences during module 3.1.

1. How has your thinking about the carbon cycle changed? What new ideas or connections did you discover about how carbon moves through the Earth's systems and how humans affect that movement?
2. What natural process did you choose to investigate next (e.g., sea level rising, coastal erosion, ocean acidification, or weather pattern changes)? How does this process affect the value your community places on Southern California's ocean?
3. What questions do you still have? What evidence, data, or models would help you understand this process better?

Module 3.2: Carbon Dioxide & Weather

Student Task: Model and Compare Natural vs. CO₂ Atmosphere Models

Model how energy from the sun flows through the atmosphere and how increased CO₂ affects heat retention, local weather, and ocean conditions. Compare a natural atmosphere to one with elevated CO₂ and include human contributions.

Section 1: Navigate - Natural Atmosphere Model

Draw your model of the natural atmosphere, or take a photo of it if you made it on a piece of paper/whiteboard:

- *Show sunlight reaching Earth.*
- *Show energy absorbed by the Earth's surface.*
- *Show energy emitted back into space.*
- *Use arrows to indicate the direction of energy flow.*

Section 2: Investigate - Elevated CO₂ Model

Draw your model of the atmosphere with elevated CO₂, or take a photo if you made it on a piece of paper/whiteboard:

- *Show sunlight reaching Earth.*
- *Show energy trapped by extra CO₂.*
- *Include human activities that produce CO₂.*
- *Use a **different color** to show trapped heat or slowed energy flow.*

Module 3.2 Day 1: Carbon Dioxide & Weather

Student Reflection

During this lesson, you explored how rising CO₂ levels in the atmosphere trap heat, change weather patterns, and affect Earth's climate. Take a moment to reflect on your experiences during module 3.2, day 1. Think about how changes in atmospheric temperature and weather patterns could affect your community and the values it holds related to Southern California's ocean.

1. How might trapped heat and changes in weather patterns impact things your community cares about, like beaches, fishing, or wildlife? Consider both **direct effects** (e.g., warmer ocean water, stronger storms) and **indirect effects** (e.g., changes in recreation, local economy, or ecosystems).

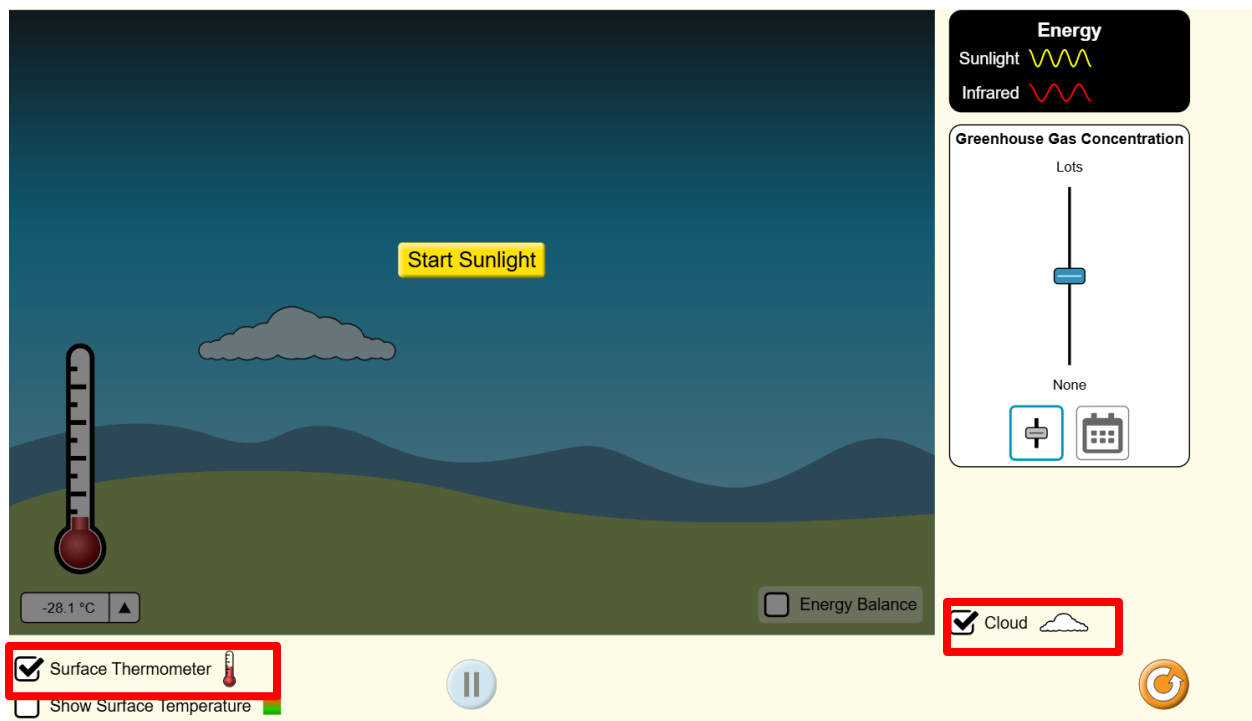
Module 3.2 Day 2: The Greenhouse Effect with PhET Simulations

Exploring Wave Interactions

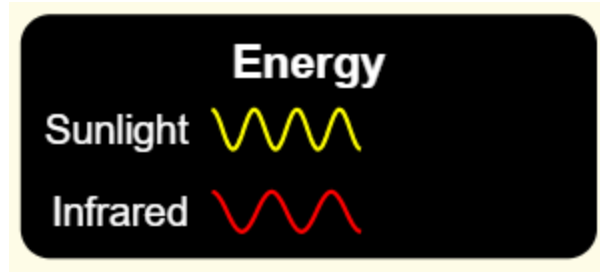
Student Instructions

Follow these steps with your team and use the PhET simulation to explore how light and infrared rays are impacted by greenhouse gas concentrations.

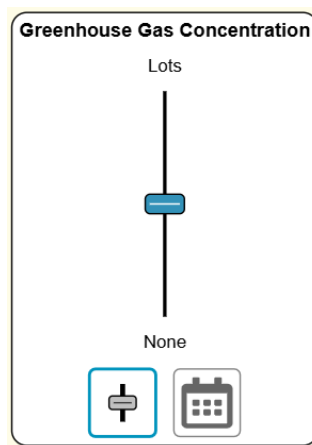
1. Visit the [PhET lab website](https://phet.colorado.edu/en/simulations/choose) and click on the “**Waves**” simulation. Before you start the simulation, make sure that you have “**Cloud**” and “**Surface Temperature**” are toggled on. Your screen should look like the following:



2. Take a moment and observe the “**Energy**” key at the top right of the screen. During the simulation you will see these two energy waves that would normally be invisible.



3. When your team is ready, start the simulation by pressing the **“Start Sunlight”** button on your screen and take 30 seconds to observe what is happening on the simulation.
 - What did you and your team observe?
 - How are energy waves interacting with the system?
4. With your simulation still running, locate the **“Greenhouse Gas Concentration”** bar. It will automatically be set to the center.



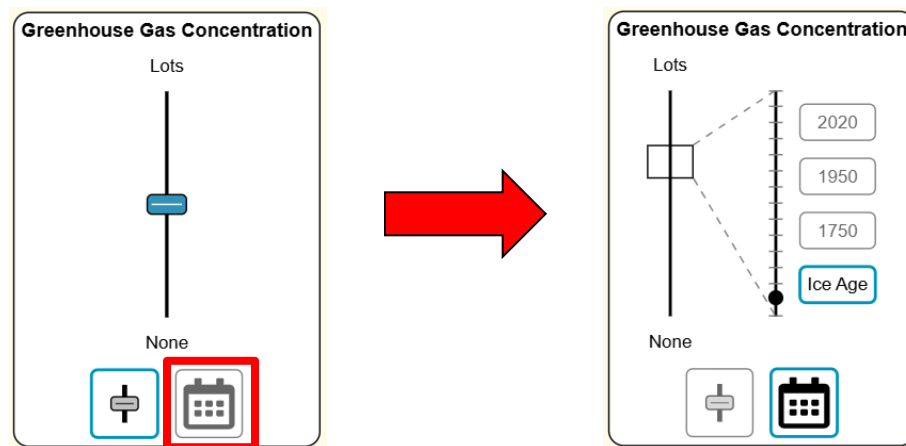
5. Take a moment to manipulate the simulation by changing the **“Greenhouse Gas Concentration”** to a higher and lower amount and observe what happens in your system.
 - How did your system change as you decreased the greenhouse gas concentration in your simulation? Why do you think this happened?
 - How did your system change as you increased the greenhouse gas concentration in your simulation? Why do you think this happened?

Module 3.2: The Greenhouse Effect with PhET Simulations

Exploring Scenarios

Now that your team has explored wave interactions on the system, take a moment to explore how the levels of greenhouse gas concentration have changed throughout the years.

1. Under the **“Greenhouse Gas Concentration”** box, click on the **calendar icon** to switch the view to include different scenarios. Your icon will change to look like the following:



When you click on the calendar icon, you will notice that your system has changed as well.

2. Next you will explore how greenhouse gas concentration has changed at different time periods and see how it has affected your system. To begin, click on **“Ice Age”** first and make initial observations on the greenhouse gas concentration and the surface temperature.
 - How would you describe the system?
 - How are energy waves interacting with the system? What impact are they having?

3. After you've made your observations for **"Ice Age"** click on the next time period, **"1750"**. With your group, answer the following questions:
 - How has your system changed from the previous time period? What is the same? What is different?
 - How would you describe the greenhouse gas concentration during this time period? What is contributing to this concentration?
 - How has your surface temperature changed between the two time periods?

4. Once you have made observations on the **"1750"** time period, repeat the process above and make observations for the **"1950"** and **"2020"** time periods.

5. Once you have observed all time periods, take a moment to answer the following with your group.
 - How has the greenhouse gas concentration changed throughout the time periods?
 - What do you think is contributing to the change of greenhouse gases?
 - What impact have these greenhouse gases had on Earth?
 - What do you predict will happen to Earth's surface temperature if these greenhouse gases continue to rise?

Module 3.2 Day 2: Carbon Dioxide & Weather

Student Reflection

During this lesson, you explored how rising CO₂ levels in the atmosphere trap heat, change weather patterns, and affect Earth's climate. Take a moment to reflect on your experiences during module 3.2, day 2. Think about how changes in atmospheric temperature and weather patterns could affect your community and the values it holds related to Southern California's ocean.

1. What did you learn today? How has your thinking changed?
2. How might trapped heat and changes in weather patterns impact things your community cares about, like beaches, fishing, or wildlife?

Module 3.3: CO₂ and Sea Level Rise/Coastal Erosion

Day 1: Investigating Sea Level Rise

Student Reflection

During this lesson, you explored how rising CO₂ in the atmosphere can trap heat, change weather patterns, and contribute to sea level rise and coastal erosion. You also connected these changes to global warming, melting ice caps, and thermal expansion of water. Take a moment to reflect on your experiences during module 3.3, day 1. Think about how changes in atmospheric temperature, storms, and sea levels could affect your community and the things it values about the Southern California ocean.

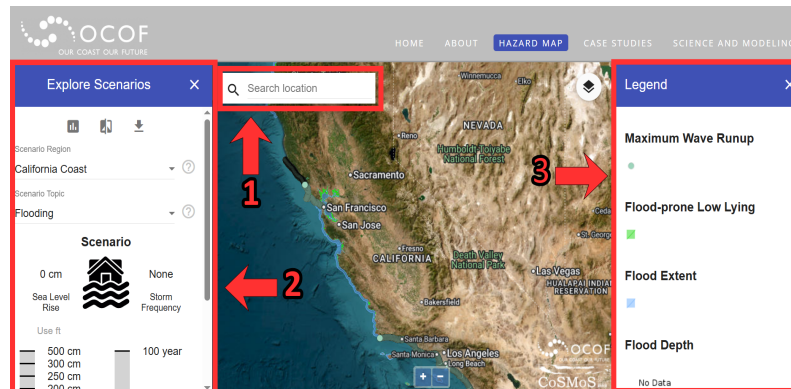
1. How has your thinking about what causes sea level rise changed? Think about new ideas or surprises you noticed while creating or revising your model.
2. How could sea level rise impact communities and things they value about the Southern California ocean?
3. Which parts of your model do you still have questions about?

Module 3.3: Impact of Sea Level Rise on Flooding in Alamitos Bay

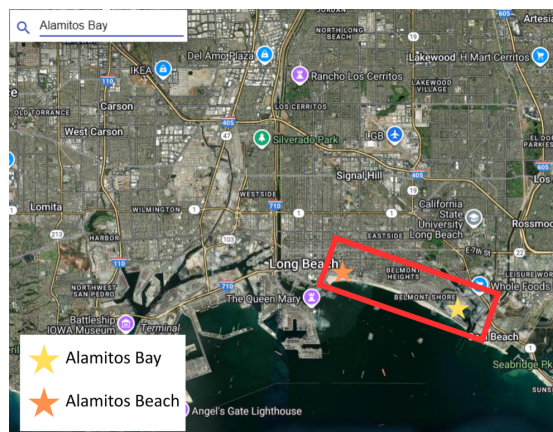
Our Coast Our Future Instructions

Follow these steps with your team and use the Our Coast Our Future simulation to explore how flooding is impacted by sea level rise.

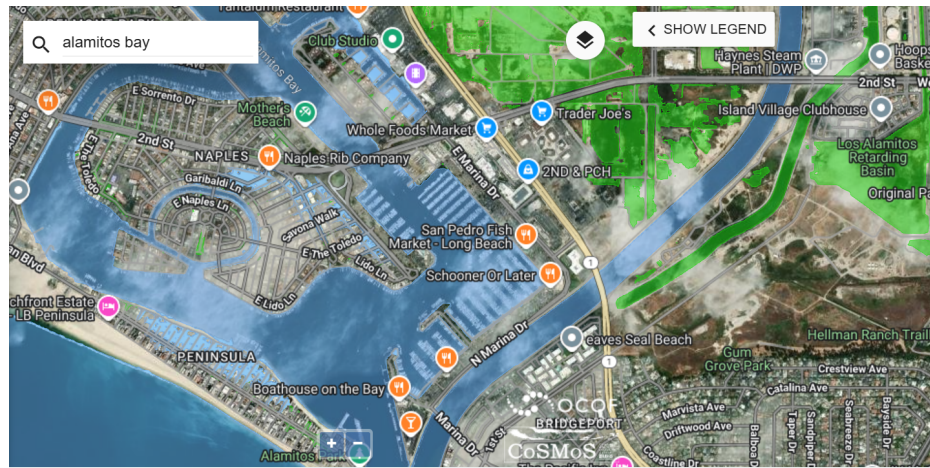
1. Visit the [Our Coast Our Future website](#). There are a couple of sections that are useful for your investigation.
 1. **Search box** is where you can search locations that the map will display.
 2. **Explore Scenarios Menu** allows you to set specific conditions for scenarios.
 3. **Legend Key** provides information about how to interpret map features using symbols.



2. In the **search box**, type in and navigate to “**Alamitos Bay**”. Our investigation will focus on two sections of the map below, Alamitos Bay and Alamitos Beach.



- Move the screen so that you have a better view of Alamitos Bay, like in the image below.



- The first scenario you will explore is how sea level rise impacts flooding in Alamitos Bay. In the **Explore Scenarios** menu, make sure that the “Scenario Topic” is ‘**Flooding**’ and “Sea Level Rise” is set to “0 cm”.

Explore Scenarios

Scenario Region

California Coast

Scenario Topic

Flooding

Scenario

0 cm

Sea Level Rise

Use ft

500 cm

300 cm

250 cm

200 cm

175 cm

150 cm

125 cm

100 cm

75 cm

50 cm

25 cm

0 cm

100 year

20 year

Annual

None

- On the right hand side menu, use the legend to identify a few features that can help you make observations of Alamitos Bay:
 - Flood-prone low lying areas** are locations that are susceptible to flooding.

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- **Flood extent** shows specific areas covered by water during a flood event.
 - **Flood depth** indicates how high water rises over land during a flood event.
6. As a group take a moment to explore Alamitos Bay and answer the following questions:
- How is flooding playing a role in Alamitos Bay with 0 cm of sea level rise?
 - How does this flooding impact the community?
7. After you've made your observation, take a moment to make a prediction with your group on how you think sea level rise might impact spaces like Alamitos Bay.
- How would sea level rise impact flooding? What impacts would it have on Alamitos Bay?
8. Now it's time to test your prediction. Move to your "**Explore Scenarios**" section and change **sea level rise** from "0 cm" to "25 cm", a projection that is estimated for 2050.
9. Once you've changed the sea level rise projection, with your group observe the same space in Alamitos Bay and discuss the following questions:
- How has the flood extent changed in Alamitos Bay between "0 cm" and "25 cm"? What is the same? What is different?

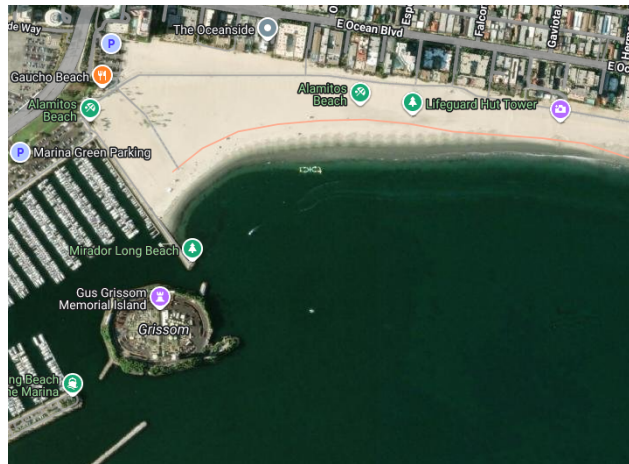
- What impact would this flooding have on the community living or recreating near Alamitos Bay? How might it impact the values communities place on the ocean?
- Was your prediction supported?

10. Once you've made your observations, take a moment to explore an even further prediction to 2100. In the "**Explore Scenarios**" section, change the **sea level rise** to "50-100" cm for the 2100 projection and make observations with your group.

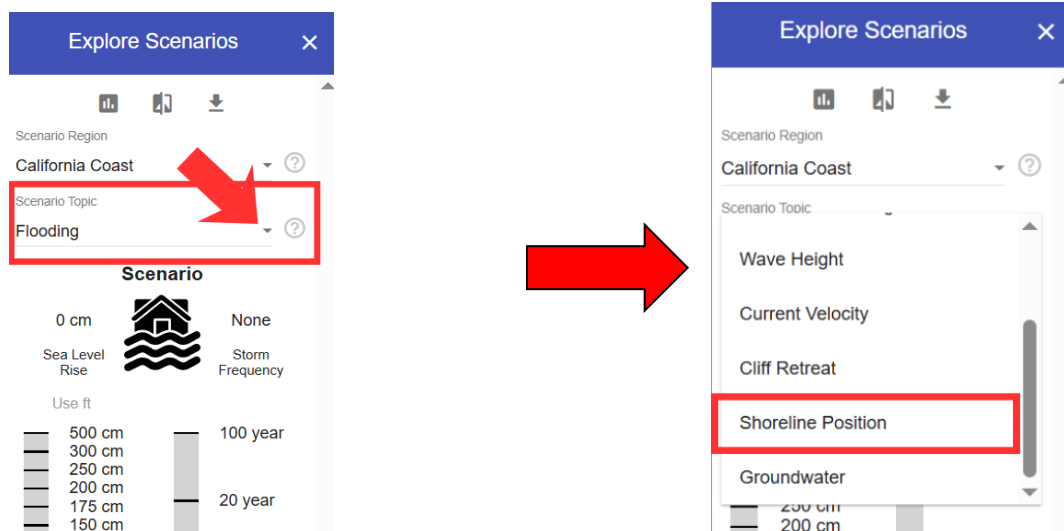
- How does the flood extent change in Alamitos Bay as sea levels are projected to rise?
- What impact would this flooding have on the community living or recreating near Alamitos Bay? How might sea level rise impact the values communities place on the ocean?

Module 3.3: Sea Level Rise's Impact on Shoreline Position on Alamitos Beach

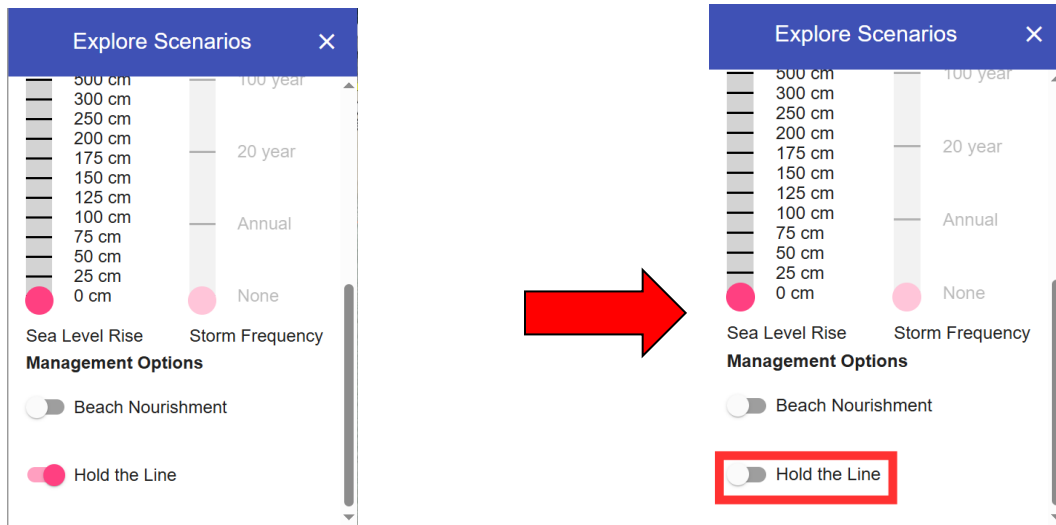
1. In the search box, type in and navigate to **“Alamitos Beach”**. Zoom into the beach area. It should look like the following:



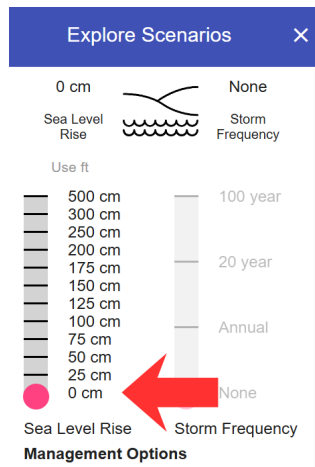
2. In the **‘Explore scenarios’** menu, navigate to the dropdown menu under **‘Scenario Topic’**. Scroll down and select **‘Shoreline Position’**.



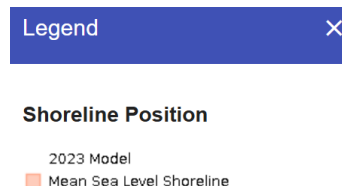
- Next, scroll down to the very bottom of the **‘Explore Scenarios’** menu and locate the option that says **“Hold the Line”**. Deselect the button next to “Hold the Line” so the bar is grey and no longer active.



- Navigate to the **“Explore Scenarios”** menu and make sure the **sea level rise** is set to **“0 cm”**.



- On the legend on the right-hand side of the screen you will see **“Mean Sea Level Shoreline”** which will indicate where the shoreline is on your simulation.



- As a group, take a moment to explore Alamitos Beach and answer the following questions:
 - How is the shoreline position playing a role in Alamitos Beach with 0 cm of sea level rise?

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- How has the shoreline position changed in Alamitos Beach between “0 cm” and “25 cm”? What is the same? What is different?
- What impact would this shoreline position have on the community living or recreating near Alamitos Beach? How might it impact the values communities place on the ocean?
- Was your prediction supported?

10. Once you’ve made your observations, take a moment to explore an even further prediction to 2100. In the “**Explore Scenarios**” section, change the **sea level rise** to “50-100” cm for the 2100 projection and make observations with your group.

- How does the shoreline position change in Alamitos Beach as sea levels are projected to rise?
- What impact would this shoreline position have on the community living or recreating near Alamitos Beach? How might sea level rise impact the values communities place on the ocean?

Module 3.3: Impacts of Sea Level Rise on Coastal Communities

Planning to Update Our SageModeler Model

After your team has explored how sea level rise may impact coastal communities using the Our Coast Our Future, take a moment to think about how you may modify your SageModeler model to reflect your newfound knowledge.

- 1. With your team take a moment to review the model you created in the first lesson and the information you gathered during the simulation exploration.
- 2. As you review, create a plan to update your Sagemodeler model using the table below.

What investigation did you take part in?	How did your thinking change as a result of this investigation?	What do you want to add, remove, or update in your SageModeler model?

Module 3.3: CO₂ and Sea Level Rise/Coastal Erosion

Day 2: Investigating Sea Level Rise

Student Reflection

During this lesson, you explored how increasing carbon dioxide (CO₂) in the atmosphere contributes to global warming and leads to sea level rise and coastal erosion. You investigated the causes of sea level rise, including melting ice caps and the thermal expansion of warming ocean water, and used models to explain these processes. You also examined real-world coastal events, such as king tides, to understand how rising sea levels can impact coastal communities and the things they value about the Southern California ocean. Take a moment to reflect on your experiences during Module 3.3, Day 2.

1. What new information or evidence changed your thinking about what causes sea level rise or how it affects communities?
2. What questions do you still have about sea level rise?
3. What updates do you plan to make to your SageModeler in the next lesson to strengthen your model?

Module 3.4 Ocean Acidification Exploration

Student Task

Basic Solution: Start of Experiment

In the space below, draw what you observe happening at the start of your experiment. Remember to label your drawing and describe any processes that are happening.

Day 1. Make a Prediction: What do you think will happen to the shell after a day in the basic solution?

Basic Solution: Day 2 of Experiment

In the space below, draw how your experiment changed after a day. Remember to label your drawing and describe any processes that are happening.

Day 2. Share your Findings: Was your prediction supported? Why or why not?

Acidic Solution: Start of Experiment

In the space below, draw what you observe happening at the start of your experiment. Remember to label your drawing and describe any processes that are happening.

Day 1. Make a Prediction: What do you think will happen to the shell after a day in the acidic solution?

Acidic Solution: Day 2 of Experiment

In the space below, draw how your experiment changed after a day. Remember to label your drawing and describe any processes that are happening.

Day 2. Share your Findings: Was your prediction supported? Why or why not?

Module 3.4: Ocean Acidification

Day 1: Student Reflection

During this lesson, you explored how increased CO₂ changes ocean chemistry and affects marine organisms that guided an updated plan to improve your SageModeler model. Take a moment to reflect on your experiences during module 3.4.

1. What new information or evidence changed your thinking about how CO₂ affects ocean ecosystems?

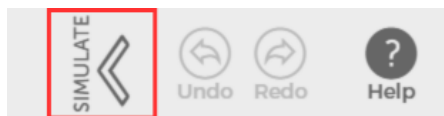
2. What questions do you still have about ocean acidification?

Module 3.4: Impact of Carbon Dioxide Emissions on the Ocean's pH and foodweb

Day 2: Sage Modeler Instructions

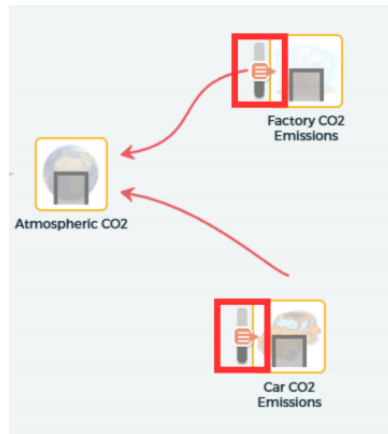
Follow these steps with your team and use SageModeler to explore how the ocean's pH and foodweb is impacted by carbon dioxide emissions.

1. Visit the [SageModeler](#) website. The first scenario you will explore is how sea water is impacted by carbon dioxide emissions.
2. As a group, take a moment to make observations of the model. In particular, view the connections to the icons and their potential relationships. Answer the following questions:
 - How does carbon dioxide get introduced into the Earth's system in the model?
3. After you've made your observations, take a moment to make a prediction with your group on how you think increasing carbon dioxide emissions might impact the ocean.
 - How might increases to these sources of carbon dioxide impact the Earth's system?
4. Once you have made your prediction, it is time to test them. To test your prediction, click on the "**Simulate**" menu in the upper right-hand corner of the platform. You'll notice that your model in the workspace changes.



5. You'll notice that sliding bars appear next to **Factory CO₂ Emissions** and **Car CO₂ Emissions**. These are variables that you can modify. The gray bars signify the measurement of its

respective variable.



6. Take a moment to manipulate the simulation to make the ocean's pH more basic (higher ocean pH).
 - What do you notice about the system when the ocean's pH is more basic?
 - How do basic conditions in the ocean's pH impact the organisms in the food web?

7. Now take to make the ocean's pH more acidic by adjusting your model (lower ocean pH).
 - What do you notice about the system when the ocean's pH is more acidic?
 - How does making the conditions more acidic in the ocean's pH impact the organisms compared to basic conditions?

8. What differences do you notice between increasing and decreasing the amount of carbon dioxide and the impacts it has on the living organisms in this model?

Module 3.4: Ocean Acidification

Planning to Update Our SageModler Model

After your team has explored how sea level rise may impact coastal communities using the Our Coast Our Future, take a moment to think about how you may modify your SageModeler model to reflect your newfound knowledge.

- 3. With your team take a moment to review the model you created in the first lesson and the information you gathered during the simulation exploration.
- 4. As you review, create a plan to update your Sagemodeler model using the table below.

What investigation did you take part in?	How did your thinking change as a result of this investigation?	What do you want to add, remove, or update in your SageModeler model?

Module 3.4: Ocean Acidification

Day 2: Student Reflection

During this lesson, you investigated how increasing carbon dioxide (CO₂) affects the ocean's pH and impacts marine organisms and food webs. You analyzed changes in your ocean acidification experiments, used a SageModeler simulation to explore how carbon dioxide interacts with seawater, and examined how these changes affect living organisms and the people who depend on the ocean. You also planned updates to your systems model to better represent the connections between carbon dioxide emissions, ocean chemistry, and community values connected to the Southern California ocean. Take a moment to reflect on your experiences during Module 3.4, Day 2. Think about how your observations, experiments, and modeling helped you understand ocean acidification, how it affects marine ecosystems and food webs, and why these changes matter to your community and its relationship with the ocean.

1. How does ocean acidification affect the ocean food web and the people who depend on it?
2. What questions do you still have about ocean acidification?
3. What updates do you plan to make to your SageModeler in the next lesson to strengthen your model?

Module 3.5: Updating Our Models

Student Task: Plan SageModeler Update

Take a moment to work with your team to plan updates to your climate change model from Module 2 using the table below.

Steps:

- 1. **Identify your focus:** *What new ideas or evidence did you learn about that will affect your model?*
- 2. **Review your existing model:** *What components and/or relationships needs to be added or updated?*
- 3. **Decide what to revise:** *Which new processes, components, or arrows will strengthen your model?*
- 4. **Connect to community values:** *How do the changes you plan to make support the values your community cares about?*
- 5. **Record your plan:** *Complete the table to outline what your will update and why.*

What new information did we learn that should be added to our model? Why is this important to add?	Which new processes or components will we add?	Which relationship or arrows will we add or revise?	What human activities influence these processes?	How does this update connect to the values our community cares about?

Module 3.5: Updating Our Models

Student Task: Make a Prediction

Take a moment to make a prediction on how your chosen value will change as carbon dioxide emissions increase.

Prediction Sentence Frame:

*We predict that as carbon dioxide emissions increase, **[our value]** will **[describe the change]**.
This is because **[explain the process using the model as evidence]**.*

Module 3.5: Updating Our Models

Student Reflection

During this lesson, you worked with your team to update your SageModeler from Module 2, based on what you have learned about sea level rise and ocean acidification. Take a moment to reflect on your experiences during module 3.5 and think about how you would communicate your findings and ideas to others.

1. If you were explaining your updated model to someone in your community, what key idea or relationship would you highlight? Why should this idea or relationship matter to them?
2. Based on what your model now shows, what action or solution could your community take to reduce or respond to these impacts?

Science Communication Product Checklist

AiSciComm Module 4

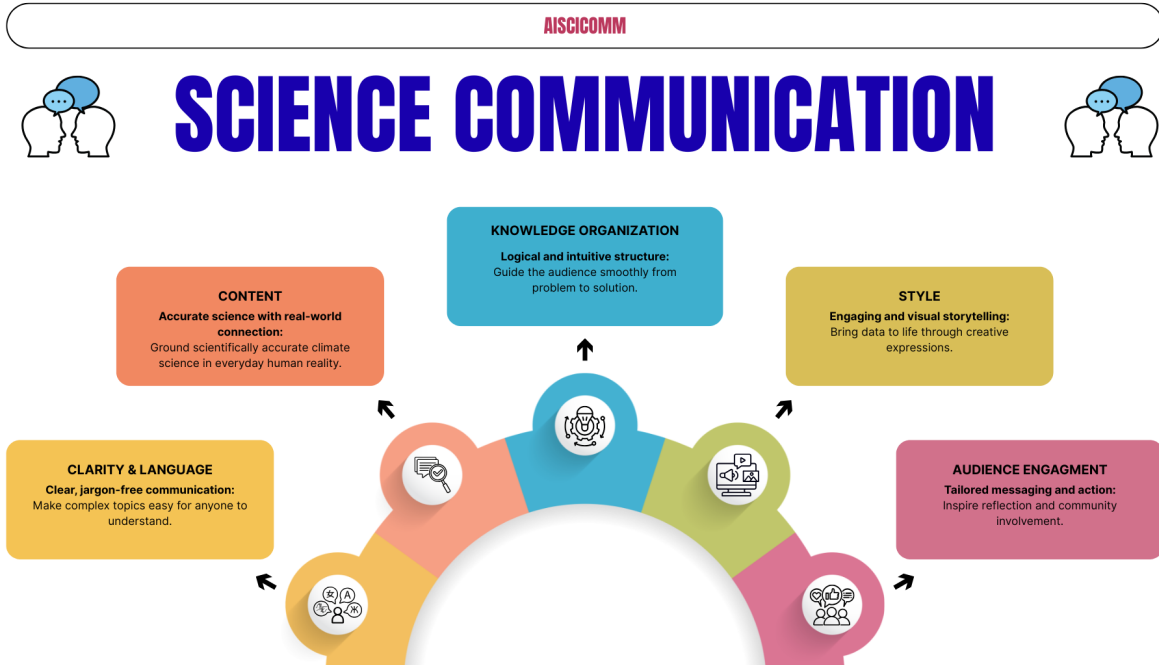
Product Task Description:	Create a science communication product explaining climate mechanisms, impacts, and solutions to a targeted audience.	
Science Communication Product Information:	Group/ Name(s): [GROUP #/ NAME(S)] Target Audience: _____ Science Communication Product Type: SELECT ONE: • Infographic • Community Event • Social Media Post (e.g., Instagram post or story) • Town Hall Presentation • Podcast • Game • Other: _____	
Science Communication Product Checklist Instructions: Use the checklist to ensure you include the following components in your science communication products.		
Component	Completion	Description
Language and Clarity	•	I explained technical terms in everyday language. <i>Did I define words like ocean acidification in context?</i>
	•	OPTIONAL: I translated key terms or phrases for my specific audience. <i>If my target audience speaks another primary language, did I include translations or accessible language?</i>
Science Accuracy	•	I checked the content to avoid scientific errors or oversimplification. <i>How do I know my content is accurate? Did I double-check where I retrieved sources of information? Did I include references to support current events or scientific facts?</i>

Real- World and Personal Connections	●	I connected climate change content to local community impacts. <i>Did I show how climate change impacts real people, economies, or wildlife? Did I connect climate change content to my community or values (reference community interviews)?</i>
	●	I included a personal connection. <i>Did I include an authentic personal story, local observation, or community perspective? Did I relate the climate change content to a local place-based reference (e.g., Orange County/ Los Angeles)?</i>
Structure	●	My product follows a clear and logical order of content. <i>Did I structure my content in an understandable flow for my targeted audience?</i>
Style and Visuals	●	I used creative elements to bring climate change content to life. <i>Did I use a narrative, metaphor, or relevant and local images to support the delivery of our message? Do all graphics, images, audio/ video clips clearly reinforce the scientific concepts being discussed?</i>
Audience Engagement	●	My tone and design fit my target audience. <i>Did I match my presentation style to my audience?</i>
	●	I included a specific way to engage my targeted audience. <i>Did I design a way to provide my audience with a concrete way to respond or reflect (e.g., Comment for Instagram post: “ Comment how your household can cut fossil fuels this week?”)</i>

Module 4.1: Identifying Solution

Student Task: Communicating Solutions

Different science communication products share key components to their audience. With your team, use the table below to identify what message you want to share with your audience/community using your models to guide your thinking.



Brainstorm: Use Key Component of Science Communication	Ideas:
Step 1: Climate Issue What specific climate issue related to the ocean is your group interested in addressing?	[Describe the climate issue in detail and its connection to your community]
Step 2: Audience Who are the possible audiences you might want to communicate this message to?	[Name the specific audience member, describe their role, and share the importance for them to hear your message]

Step 3: Message What is your message? What do you want your target audience to know or feel at the end of the message?	[Detail your message]
Step 4: Evidence What data or evidence supports your message? What should you include in your communication?	[Detail your evidence/ data collected from previous modules]
Draft and Revise Message	Ideas:
Draft: Write a 1-2 sentence version of your message.	
Audience Feedback: What did other groups/ the chatbot suggest? Do you agree/ disagree with the suggestions? Why or why not? What improvements could be made?	
Revised Message: What would be a revised message you would share?	

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Module 4.1: Identifying Solutions

Student Reflection

During this lesson, you worked with your team to draft and revise a message to an intended audience about a climate change-related issue. Take a moment to reflect on your experiences during module 4.1.

1. What did you learn about communicating your message effectively?

Consider the three science communication principles:

- a. **Intentionality:** How did thinking about your audience shape your message?
 - b. **Reciprocity:** What did you learn from testing your message and listening to feedback?
 - c. **Reflexivity:** How did you reflect on your choices and improve your message?
-
2. How did feedback change your thinking and message? Explain what you revised and why it made the message stronger or clearer.

Module 4.2: Choose Our Audience and Plan Our Approach

Student Task: Planning Your Communication Product

There are different products and platforms that can be used in science communication, such as infographics, podcasts, flyers, games, and social media posts, and that target a specific audience. With your team, plan your own science communication product. Use the table below to plan your message and communication approach. You are not creating the final product yet, this is a planning space for ideas.

Target Audience Who are you trying to reach? What do they care about? Why is it important for them to hear this message?	Ideas:
Key Message What is the main idea you want people to learn, remember, or feel?	[Recall your message from Module 4.1] - Our audience is ... because they care about ... - Our communication product will be a ... because ... - The most important message people should remember is ... - Evidence from our model that supports this message is ... - We want our audience to feel ... so they will ...
Product Type What science communication format will you use? Why is this format effective for your audience?	Select a science communication format or add your own idea: ☐ Social media posts (Instagram/TikTok) ☐ Infographic/ flyer ☐ Townhall presentation ☐ Board games ☐ Other: Why is this format effective?

Text/ Words What text or wording might you include?	[Include key phrases, slogan, call to action, short explanation]
Visuals What visuals might you include?	[Images, diagrams, icons, color choices, layout ideas]
Data/ Evidence What evidence from your model or previous lesson supports your message?	[Graphs, numbers, examples]
Emotional/ Personal Appeal How will you connect emotionally to your audience? How should they feel?	[Example emotions - concerns, hopeful, empowered, etc.]
Define Success How will you know if you are successful with your science communication plan? How will you be able to tell if your audience has learned or begun to feel the way that you want them to?	
Feedback: Note any feedback others shared about what they like and suggestions for improvement.	

Module 4.2: Choose Our Audience and Plan Our Approach

Optional: Student Reflection

During this lesson, you analyzed examples of science communication and planned how you will communicate your message about climate change impacts on the ocean to a specific community audience. Take a moment to reflect on your experiences during Module 4.2.

1. How did thinking about your audience change the way you designed your message?

2. Reflect on the science communication principles and your message:
 - a. **Intentionality:** How will you purposefully reach your audience? What will you do to find out more about your audience?

 - b. **Reciprocity:** How will you test your message and learn from feedback?

 - c. **Reflexivity:** How will you reflect and make improvements based on what you learn?

3. What questions do you still have about communicating science effectively or something you want to learn more about before creating your product?

Module 4.3: Building a Chatbot to Test Our Message

Student Task: Testing Your Science Communication Message with a Chatbot

Use a GenAI chatbot to represent your target audience and test your science communication message. Complete the table below to guide your thinking.

Steps to Complete	Ideas:
Define your chatbot audience	Role: Task: Background: [age, race/ethnicity, lived experiences, occupation] Constraints: [response length, focus on science communication]
Create your chatbot prompt	Using the information above, write a prompt that your chatbot will follow. Example format: You are role-playing as {character}. Here are details about you: you will act as this role: {role}. In this task, I want you to {provide task description}. You will role-play as an individual with this background information: {describe background, like age, race/ethnicity, living experiences, occupation}. Your response should have the following constraints: {constraints}. I will share details about a science communication product with you, and you will critique my plan from the perspective of this character. Here is my product type: {product type, like a social media post, board game, or infographic}. Here is my science communication plan: {plan}. What do you think about this plan? Ask me questions about what is unclear or can be improved about my plan.

Test your message	• Input your prompt into the AISciComm chatbot. • Share your science communication plan. • Ask the chatbot for specific feedback and note any questions it asks.
Record your results	• Take screenshots or summarize the conversation. • What suggestions or questions did your chatbot provide? • How could this feedback improve your message?
Revisions	[What revisions will you add or modify to your plan in Module 4.2]

Module 4.3: Building a Chatbot to Test Our Message

Student Reflection

During this lesson, you used a GenAI chatbot to represent your target audience and test your science communication message. Take a moment to reflect on your experiences during Module 4.3.

1. How did the chatbot respond to your message? Was there anything surprising or interesting about the feedback you received?
2. What did you learn about your audience from this conversation?
3. Did this feedback make you think differently about how to communicate your science topic? How?
4. How might this chatbot experience be similar to or different from talking to a real person? How might you validate what the chatbot shared?

Module 4.4: Put Our Plan Into Action

Student Task: Build Our Science Communication Product

With your group, create your science communication products, based on the plans and suggested revisions from Module 4.2 and Module 4.3. You may draft your products in [Canva](#).

- Canva: <https://www.canva.com/>
 - Infographic templates: <https://www.canva.com/infographics/templates/>
 - Instagram templates: <https://www.canva.com/instagram-posts/templates/>
 - Board game idea templates: <https://www.canva.com/templates/?query=board-game-interact>
- If students are creating a board game, they could use generative AI (e.g., ChatGPT, our [AISciComm Default chatbot](#)) to help them refine the rules of the game.

Upload your final product to the Google Drive and Screen shot and add the product to this page.

Module 4.4: Put Our Plan Into Action

Student Reflection

During this lesson, you created a science communication product and explored other teams' products. Take a moment to reflect on your experiences during Module 4.4.

After completing the virtual gallery walk:

1. What patterns or trends did you notice across the science communication products you reviewed?
2. Were there any ideas, visuals, or strategies from other teams that you would like to incorporate into your own product? What might you suggest to improve their messages? Be specific on which groups you were reviewing.
3. What surprised you about the different ways teams communicated the same climate issue?